# Introduction to Statistical Learning

## Review of Math Prerequisites

Image credit: 'The Mayflower at Sea' by Granville Perkins, 1876.
Wallach Division Picture Collection, The New York Public Library.

# Notation

- $a \in A$ *set membership*: *a* is *member* of set *A*
- $|B|$ *cardinality*: number of items in set *B*
- $\| \mathbf{v} \|$ *norm*: "length" of vector *v*
  - *We'll use Manhattan/L1 & Euclidean/L2 distances*
- $\sum$ *summation*
- $\int$ *integral (think: summation)*
- $\mathbb{R}$, the set of *real* numbers
- $\mathbb{R}^n$ *real number space* of dimension *n*

  n = 2 : plane or 2-space
  n = 3 : 3- (dimensional) space
  n > 3 : *n*-space or *hyperspace*

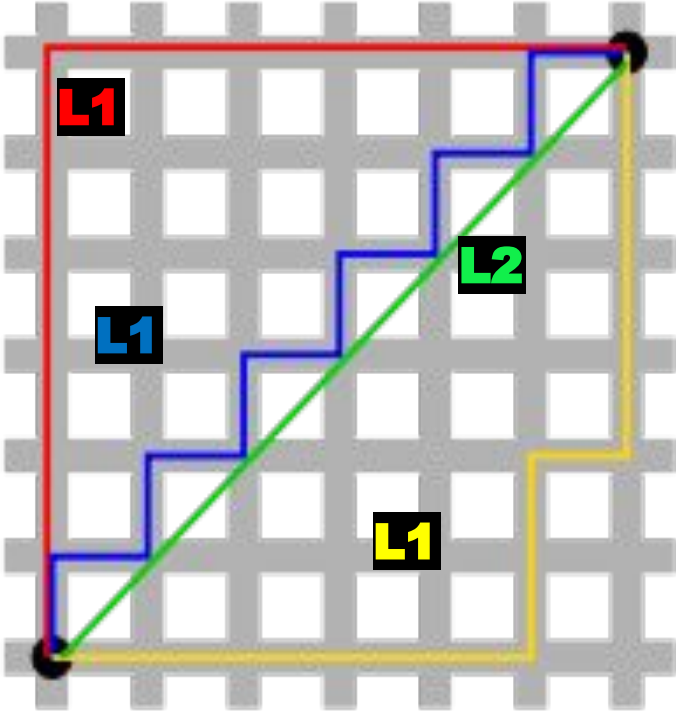





Image: https://en.wikipedia.org/wiki/Taxicab_geometry

# Notation

- $dy/dx$ *derivative* of $y$ with respect to $x$
- $f: A \to B$, a *function* (*map*) from domain $A$ to range $B$
  - $A, B$ can be subsets of multidimensional $\mathbb{R}^n$, $n > 1$
    - Eg. $f(x), f(\mathbf{x}), \mathbf{f}(x), \mathbf{f}(\mathbf{x})$
- $\frac{\partial y}{\partial x_i}$ partial derivative of $y$ with respect to element $i$ of vector **x**
- Often the **context** suggests the structure
- $X_{n \times p}$ is a $n \times p$ **data matrix**
  - columns $X_i$ as **features**, rows $x_j$ as **observations**
  - **x**, **y**, **z**, *vector* (often lower case, sometimes bold)
  - **A, B, X** *matrix* (often uppercase, sometimes bold)

# The concept of probability

**Intuition**:

- In some process, several outcomes are possible
- When the process is repeated many times, each outcome occurs with a characteristic of ***relative frequency***, or ***probability***
  - This is the observed or empirical probability
- If a particular outcome happens more often than another outcome, we say it is more probable

# The concept of probability

Arises in **theoretical** and **empirical** contexts:

- In **idealized** conceptions of a repeated process.
  - Ex: You consider the behavior of an unbiased six-sided dice. The ***expected*** probability of rolling a 5 is 1 / 6 = 0.1667.
  - Ex: You need a model for how **people's heights** are distributed. You choose a normal distribution (bell-shaped curve) to represent the ***expected*** relative probabilities.
- In **actual** repeated experiments
  - Ex: You record the color of 1000 cars driving by. 57 of them are green. You ***estimate*** the probability of a car being green as 57 / 1000 = 0.0057.

Image credit: https://en.wikipedia.org/wiki/Probability

# Probability spaces

A **probability space** is a mathematical structure with operations:

- $\Omega$, **sample space** is the set of possible **outcomes**, ω
  - number of possible outcomes $= |\Omega| = N$ (cardinality or size of a set)
- $\mathcal{F}$, **event space** where $E \in \mathcal{F}$ is a set of outcomes
  - an event comprises 0 to $N$ outcomes
  - number of possible events $= |\mathcal{F}| = 2^N$ , i.e. a **power set** of $\Omega$
- $\mathbb{P}$, ***probability distribution*** *(or measure)*
  - maps events to real numbers in $[0,1]$ (the *probability* of $E$)
  - probability of $E$ is the *sum* of probabilities of possible outcomes in $E$

- Ex: rolling a fair coin with $\Omega = \{H, T\}$
  - $\mathcal{F} = \{\{\ \},\{H\},\{T\},\{H,T\}\}$
  - $\mathbb{P}(\{\ \}) = 0$
  - $\mathbb{P}(\{H\}) = ½$
  - $\mathbb{P}(\{H,T\}) = \mathbb{P}(\{H\}) + \mathbb{P}(\{T\}) = ½ + ½ = 1$

Axioms of probability

1. **Non-negativity**:

   for any event $E \in \mathcal{F}, \mathbb{P}(E) \geq 0$

2. A **unit measure**: all possible outcomes:

   $\mathbb{P}(\Omega) = 1$

3. **Additivity of disjoint events**:

   for all (disjoint) events $E, E' \in \mathcal{F}$ where $E \cap E' = \emptyset$ we have

$$\mathbb{P}(E \cup E') = \mathbb{P}E + \mathbb{P}E'$$

Define $|\Omega|$ = number of possible outcomes

- **Discrete** space is **countable** (finite or infinite)
  - Ex. $\mathbb{N}$, space of naturals, or $\{-1,2,10\}$
  - Analysis involves *summations* of probability-weighted values
  - Ex. coin flip outcomes
  - Ex. count of customers at a store in one day

- **Continuous** space is uncountable
  - Ex. $\mathbb{R}$ space of reals
  - Analysis involves *integrals* of probability-weighted values

# Example of discrete probability space

Single roll of a six-sided dice

- 6 possible outcomes: $\omega = 1,2,3,4,5,6$
- $2^6 = 64$ possible events
  - Ex: odd outcome is $E = \{1,3,5\}$
- If die is fair, then probabilities of outcomes are equal

$$\mathbb{P}(1) = \mathbb{P}(2) = \mathbb{P}(3) = \mathbb{P}(4) = \mathbb{P}(5) = \mathbb{P}(6) = \frac{1}{6}$$

  - Ex: probability of an odd face is

$$\mathbb{P}(1) + \mathbb{P}(3) + \mathbb{P}(5) = \frac{3}{6}$$

- Read $\mathbb{P}(4)$ as $\mathbb{P}[\omega = 4]$ or $\mathbb{P}(\{4\})$, i.e. we are measuring a set

# Example of discrete probability space

Three consecutive flips of a coin

- 8 possible outcomes:
  - $\omega$ = HHH, HHT, HTH, HTT, THH, THT, TTH, TTT
- $2^8 = 256$ possible events
  - Ex: $E = (\omega \in \{ \text{HHT, HTH, THH} \})$, i.e. exactly two flips are heads
  - Ex: $E = (\omega \in \{ \text{THT, TTT} \})$, i.e. the first and third flips are tails
- If coin is fair, then probabilities of outcomes are equal

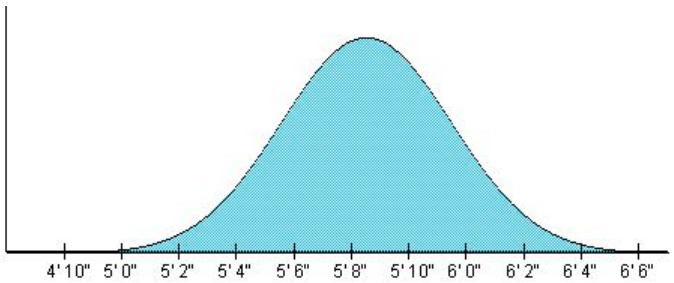
$$P(\text{HHH}) = P(\text{HHT}) = P(\text{HTH}) = P(\text{HTT})$$
$$= P(\text{THH}) = P(\text{THT}) = P(\text{TTH}) = P(\text{TTT}) = 1 / 8$$

  - Ex: probability of event $E$ = ( exactly two heads ) is
    $P(\text{HHT}) + P(\text{HTH}) + P(\text{THH}) = 3 / 8$

# Example of continuous probability space

Height of a randomly chosen male

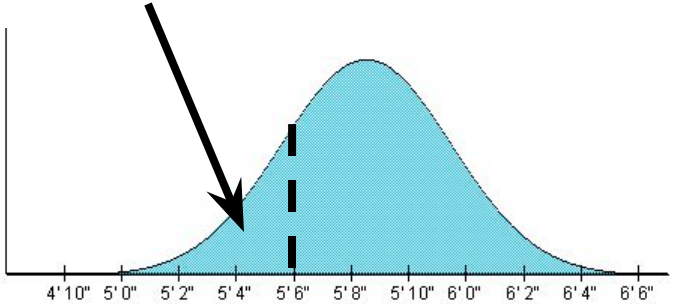
- Infinite number of possible outcomes: ω has some single value in range 2 feet to 8 feet
- Infinite number of possible events
  - Ex: $E = \{ \omega \mid \omega < 5.5 \text{ feet} \}$, i.e. individual chosen is less than 5.5 feet tall
- Probabilities of outcomes are not equal, and are described by a continuous function, $P(\omega)$



# Example of continuous probability space

Height of a randomly chosen male

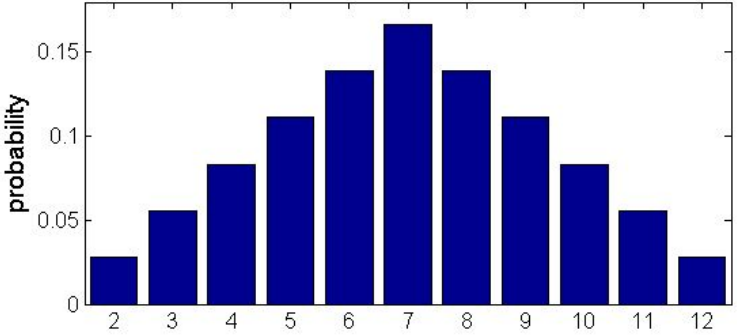
- Probabilities of outcomes ω are not equal, and are described by a continuous function, $p(\omega)$
- $P(\omega)$ is a *relative*, not an *absolute* probability
  - $P(\omega)$ for any ω is zero
  - $\int_{\mathbb{R}} P(\omega)\, d\omega = 1$ (i.e. area under curve) is 1
  - Ex: $P(\omega < 5'6'') = (\int P(\omega)\, d\omega$ from $\omega = -\infty$ to $5'6'') \approx 0.25$



Probability distributions

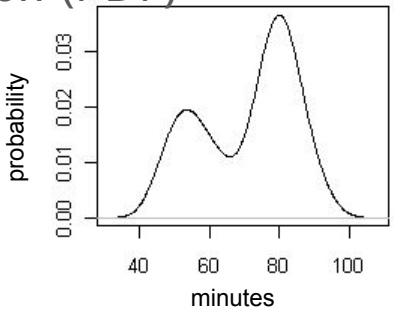
- Discrete: *probability mass function* (PMF)

Ex: sum of two fair dice




- Continuous: *probability density function* (PDF)

Ex: waiting time between
eruptions of Old Faithful geyser



# Random variables (RV)

Don't be confused with that kind of RVs:

- A RV $X$ is a function that **maps outcomes to reals**
  - Common notation: $X(\omega) = x$, or just $X = x$
  - $X$ can be discrete or continuous
- It helps redefine (simplify) a probability space to a new probability space
- Ex: $X$ = # of heads in three flips of a coin. $X \in \{0, 1, 2, 3\}$
  - $p(X = 0) = p(X = 3) = 1 / 8$
  - $p(X = 1) = p(X = 2) = 3 / 8$
  - Size of space (# of "outcomes") reduced from 8 to 4

Image credit: Breaking Bad crime drama TV series

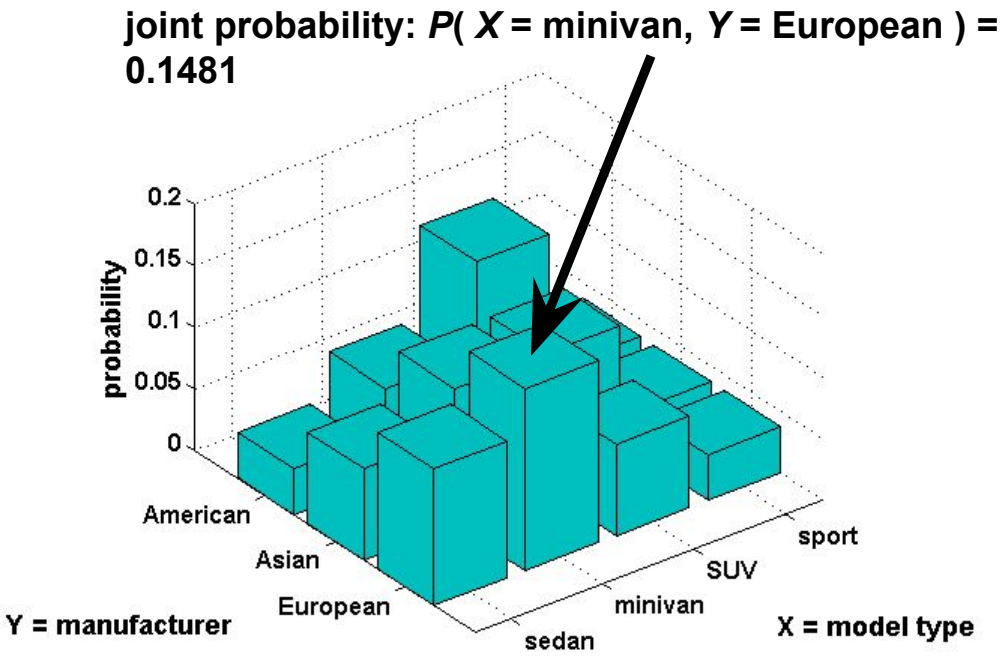
# Multivariate probability distributions

- Scenario
  - Several random processes occur
    - doesn't matter whether in parallel or in sequence
  - Want to know probabilities for each possible combination of outcomes

- Can describe as **joint probability** of several RVs
  - Ex: 2 processes whose outcomes are represented by RVs *X* and *Y*. Probability that process *X* has outcome *x* <u>and</u> process *Y* has outcome *y* is denoted as:

$$P(X = x, Y = y)$$

# Example of multivariate distribution



# Multivariate probability distributions

- **Marginal** probability
  - Probability distribution of a single variable in a joint distribution
  - Ex: 2 RVs $X$ and $Y$:

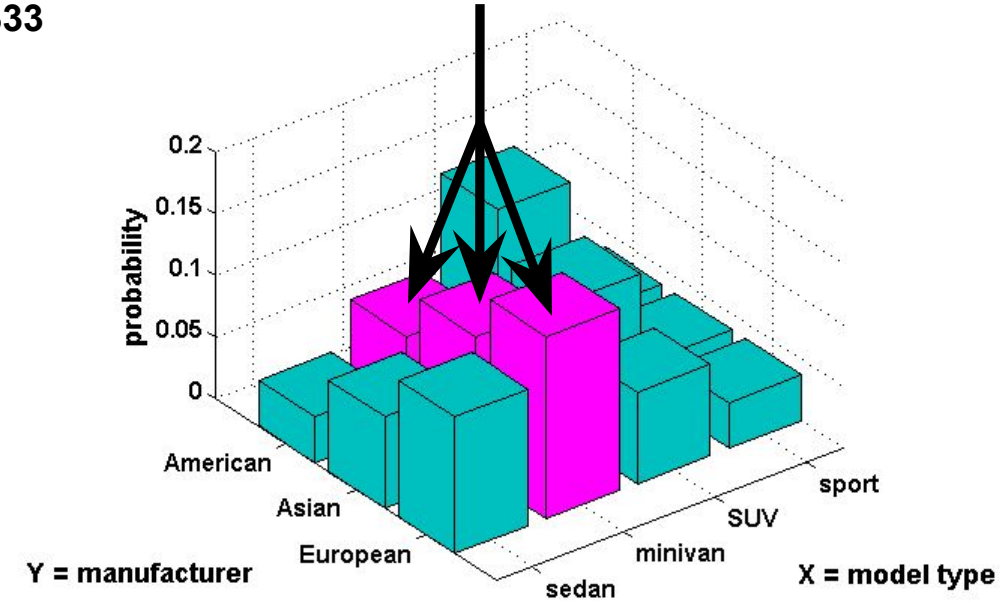
    $$P(X = x) = \sum_{b=\text{all values of } Y} P(X = x, Y = b)$$

- **Conditional** probability
  - Probability distribution of one variable *given* that another variable takes a certain value
  - Ex: 2 RVs $X$ and $Y$.

    $$P(X = x \mid Y = y) = P(X = x, Y = y) / P(Y = y)$$

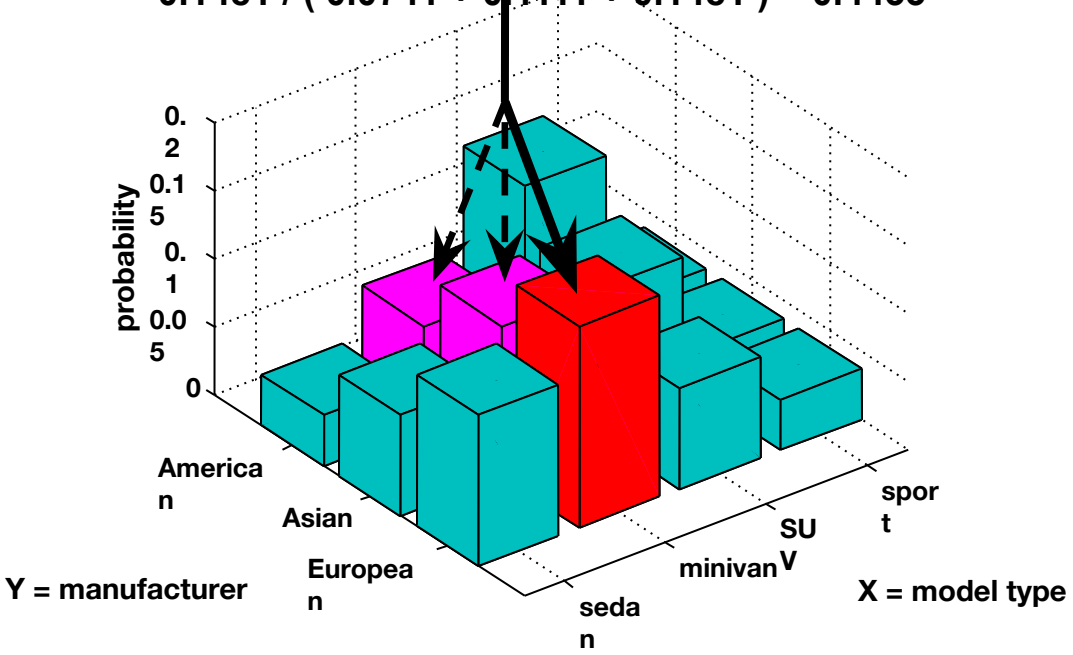
# Example of marginal probability

**marginal probability: $P(X = \text{minivan}) = 0.0741 + 0.1111 + 0.1481 = 0.3333$**



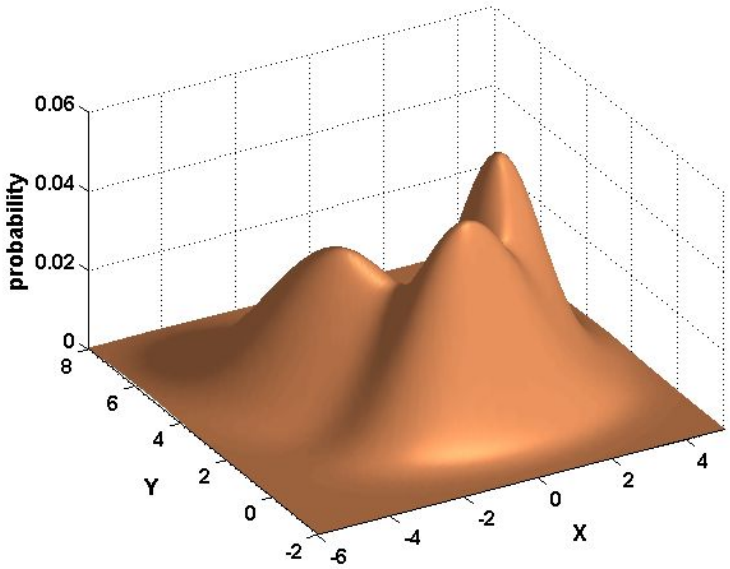
# Example of conditional probability

**conditional probability: $p(\ Y = \text{European} \mid X = \text{minivan}\ )$**
**=**
**0.1481 / ( 0.0741 + 0.1111 + 0.1481 ) = 0.4433**



# Continuous multivariate distribution

- Same concepts of joint, marginal, and conditional probabilities apply (except use integrals)
- Ex: 3-component Gaussian mixture in 2 dimensions



# Expected value

Given:

- A discrete RV $X$, with possible values $x = x_1, x_2, \ldots x_n$
- Probabilities $p( X = x_i )$ that $X$ takes on the various values of $x_i$
- A function $f$ defined on range of $X$

The **expected value** of $f$ is the **probability-weighted "average"** value of $f( x_i )$:

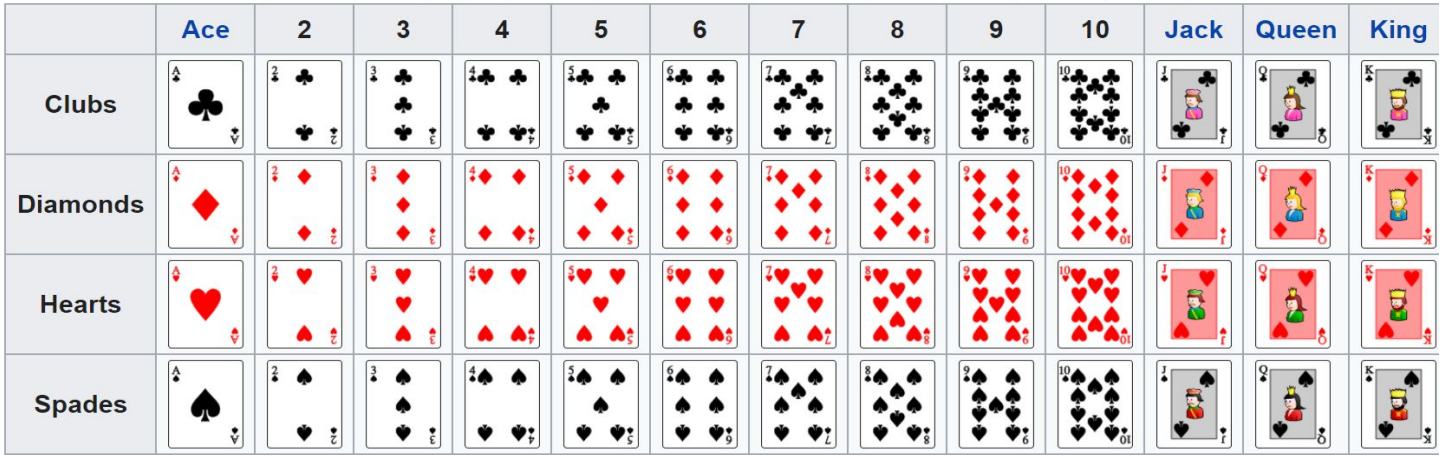
$$E( f ) = \sum_i P( x_i ) \cdot f( x_i )$$

# Example of expected value

- We draw a card from a deck
  - If face card (F), you get \$10; else (N) you pay \$4
- RV $X \in \{F, N\}$. $P(F) = 3/13$; $P(N) = 10/13$
- $f(X)$ is the payout: $f(F) = 10$; $f(N) = -4$
- **Expected** payout is:
  - $E(f) = \sum_i P(x_i) \cdot f(x_i) = 3/13 \cdot \$10 + 10/13 \cdot \$-4 = \$-0.77$

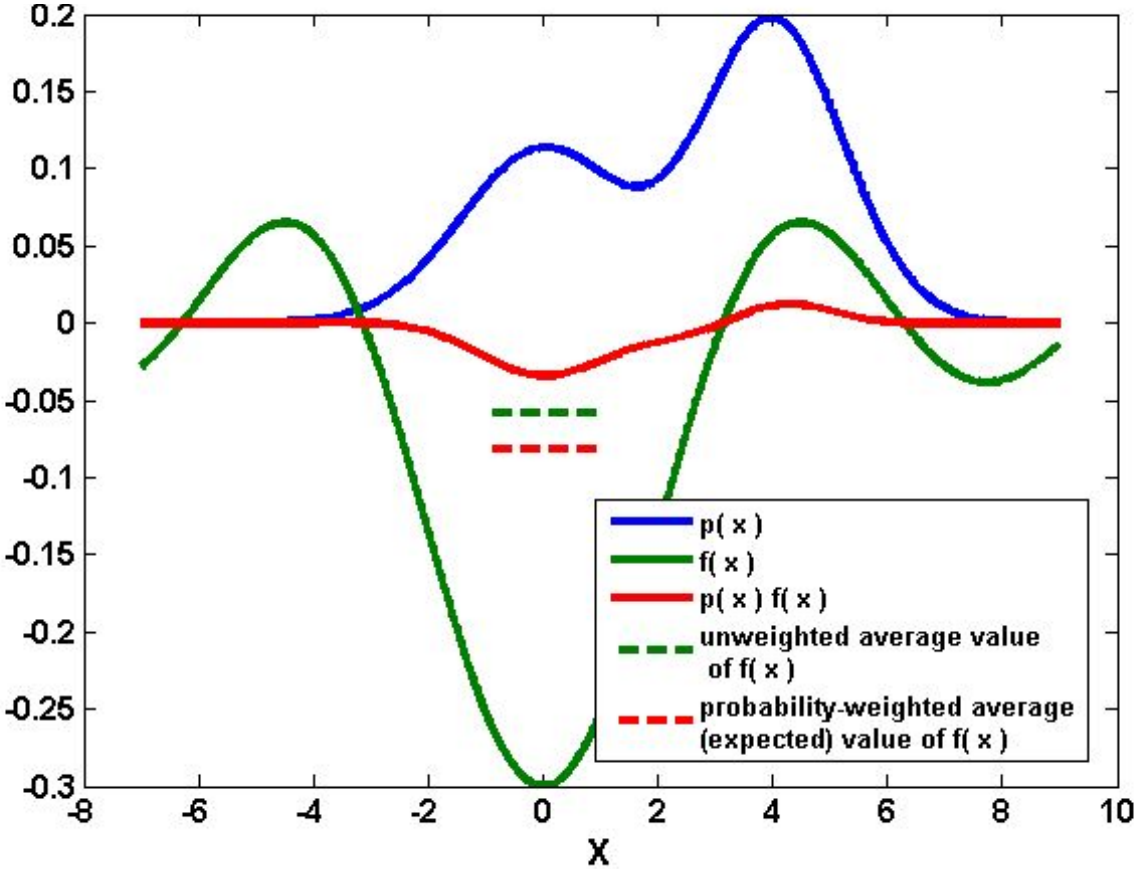
Image credit:
https://en.wikipedia.org/wiki/Standard_52-card_deck

**Example set of 52 playing cards; 13 of each suit clubs, diamonds, hearts, and spades**



# Expected value in continuous spaces

$$E(f) = \int_{x \in [a,b]} p(x) \cdot f(x)\, dx$$



# Common forms of expected value (1)

- Mean ($\mu$)

  $f(x_i) = x_i \quad \Rightarrow \quad \mu = E(f) = \sum_i p(x_i) \cdot x_i$

  - Average value of $X = x_i$, taking into account probability of the various $x_i$
  - Most common measure of **"center" of a distribution**

- Compare to formula for a **sample mean**
  - Weights are equal

$$\mu = \frac{1}{N} \sum_{i=1}^{n} x_i$$

# Common forms of expected value (2)

- Variance ($\sigma^2$)

  $f(x_i) = (x_i - \mu) \quad \Rightarrow \quad \sigma^2 = \sum_i p(x_i) \cdot (x_i - \mu)^2$

  - **Average value of squared deviation** of $X = x_i$ from mean $\mu$, taking into account probability of the various $x_i$
  - Most common measure of "spread" of a distribution
  - $\sigma$ is the *standard deviation*

- Compare to formula for **sample variance**

$$\sigma^2 = \frac{1}{N-1}\sum_{i=1}^{n}(x_i - \mu)^2$$

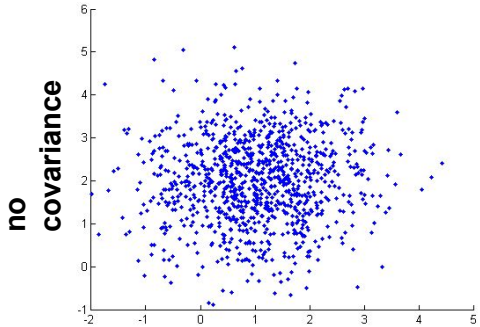
# Common forms of expected value (3)

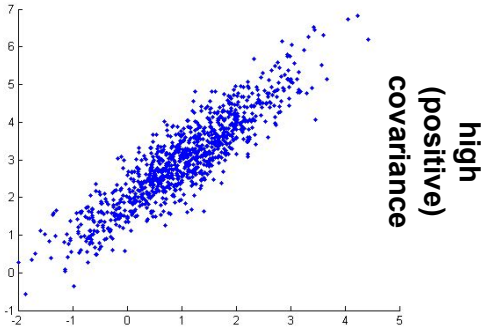
- Covariance

  $f(x_i) = (x_i - \mu_x), \quad g(y_i) = (y_i - \mu_y) \quad \Rightarrow$

  $\mathrm{cov}(x, y) = \sum_i p(x_i, y_i) \cdot (x_i - \mu_x) \cdot (y_i - \mu_y)$

  - Measures tendency for *x* and *y* to deviate from their means in same (or opposite) directions at same time




- Compare to formula for covariance of actual samples

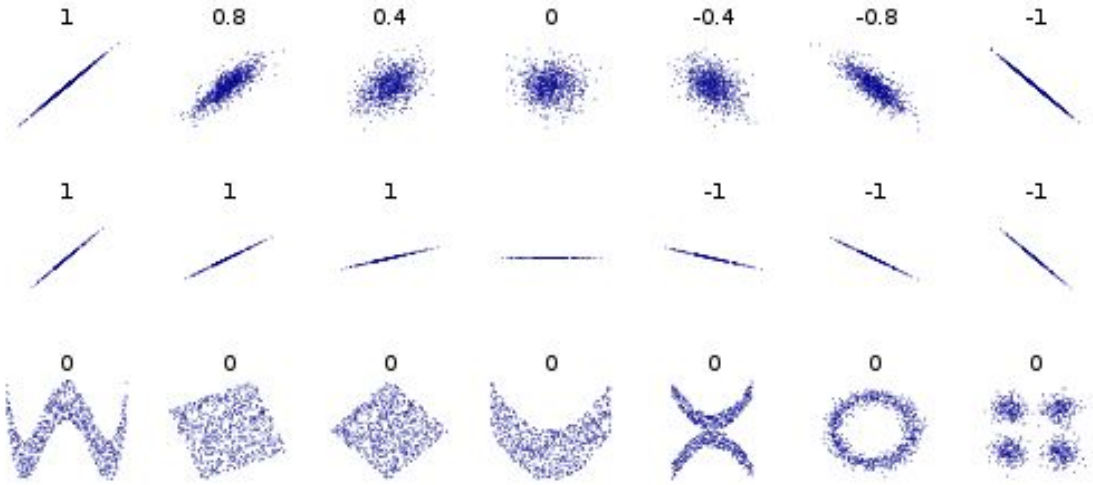
$$\mathrm{cov}(x, y) = \frac{1}{N-1} \sum_{i=1}^{n} (x_i - \mu_x)(y_i - \mu_y)$$

# Correlation

- Pearson's correlation coefficient is covariance normalized by the standard deviations of the two variables

$$\mathrm{corr}(x, y) = \frac{\mathrm{cov}(x, y)}{\sigma_x \sigma_y}$$

  - Always lies in range -1 to 1
  - Only reflects *linear dependence* between variables




Linear dependence with noise

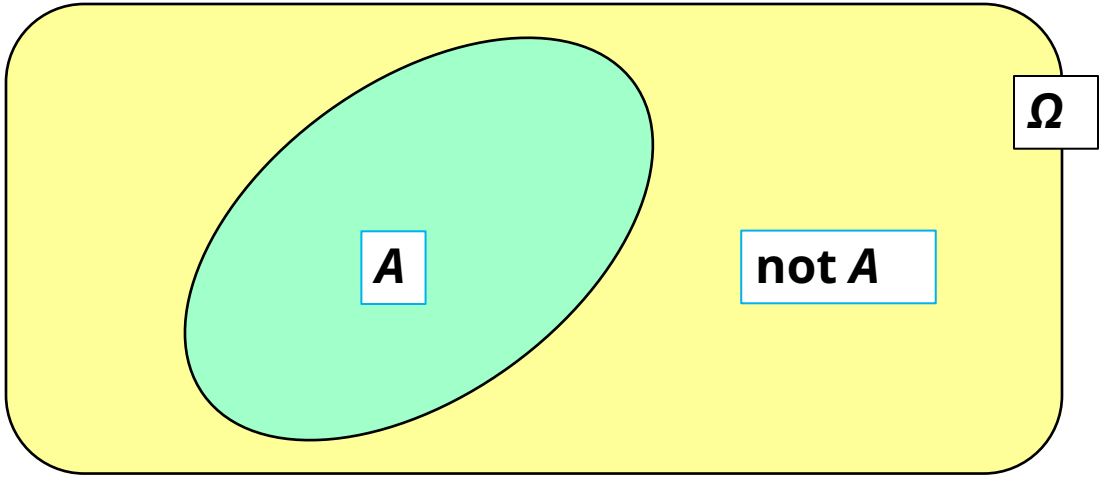
Linear dependence without noise

Various nonlinear dependencies

# Complement rule

Given: event *A*, which can occur or not

$$P(\text{ not } A) = 1 - P(A)$$



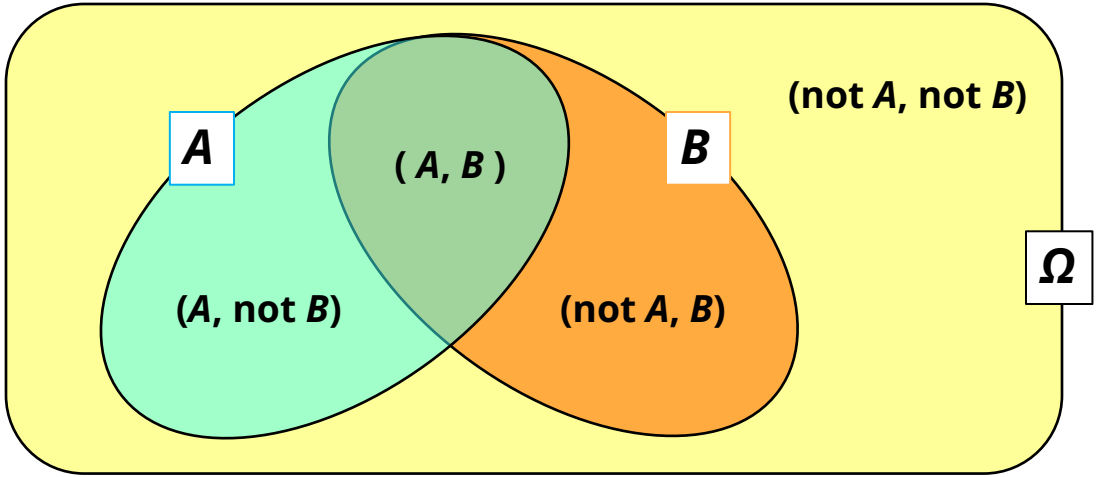

areas represent relative probabilities

# Product rule

Given: events $A$ and $B$, which can co-occur (or not)

$$P(A, B) = P(A \mid B) \cdot P(B)$$

(same expression given previously to define conditional probability)




areas represent relative probabilities

# Example of product rule

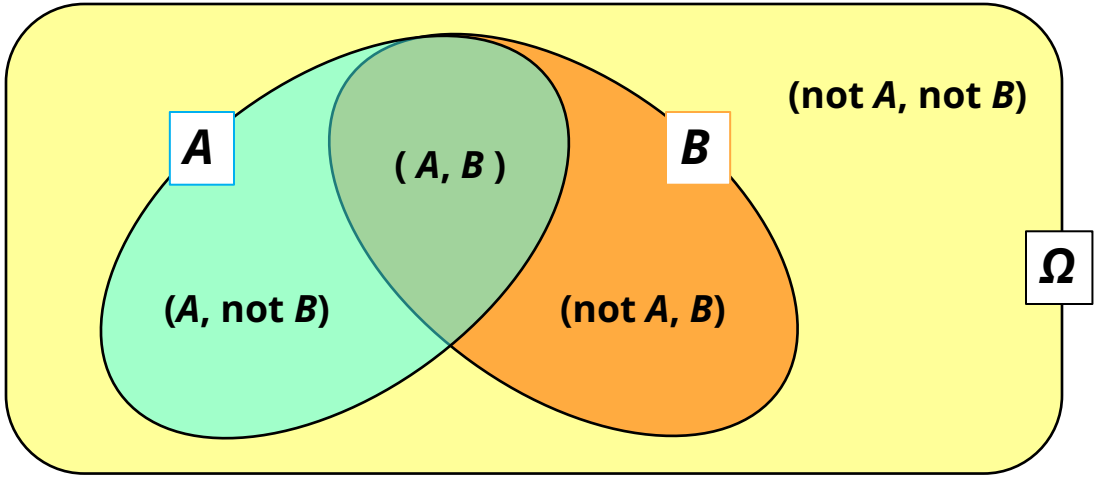
- Probability that a man has white hair (event *W*) <u>and</u> age over 65 (event *A*)
  - $P(A) = 0.18$
    - 18% chance of picking a man with white hair
  - $P(W \mid A) = 0.78$
    - Among all 65+ men, 78% have white hair
  - $P(W, A) = P(W \mid A) \cdot P(A) = 0.78 \cdot 0.18 = 0.14$
    - 14% chance of picking a man with white hair and >65

# Rule of total probability

Given: events $A$ and $B$, which can co-occur (or not)

$$P(A) = P(A, B) + P(A, \text{not } B)$$

(same expression given previously to define marginal probability)

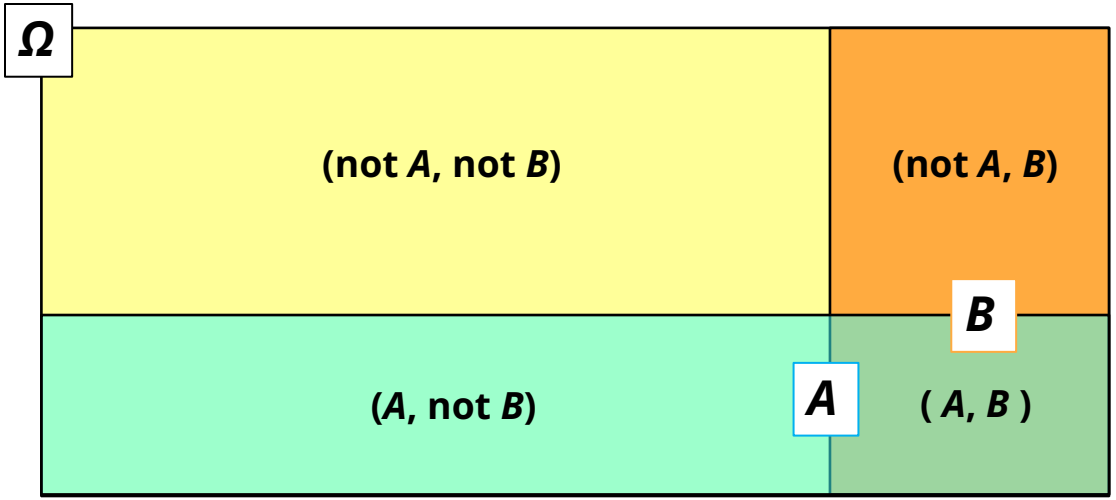



areas represent relative probabilities

# Independence

Given: events $A$ and $B$, which can co-occur (or not)

$$p(A \mid B) = p(A) \quad \text{or} \quad p(A, B) = p(A) \cdot p(B)$$




areas represent relative probabilities

# Examples of independence / dependence

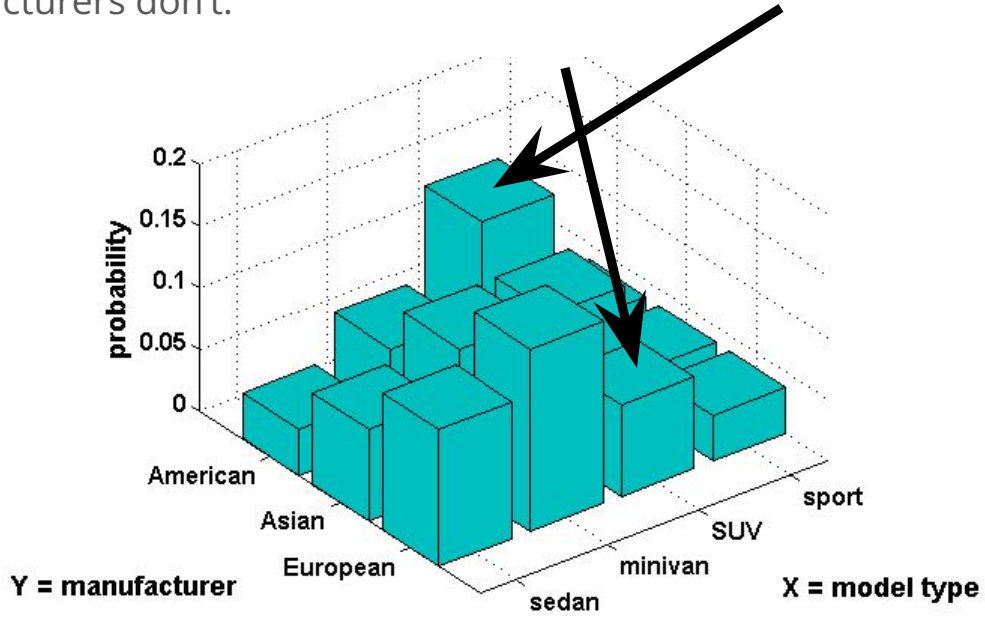
- Independence:
  - Outcomes on multiple rolls of a dice
  - Outcomes on multiple flips of a coin
  - Height of two *unrelated* individuals
  - Successively drawing a king from a deck, if card from each draw is ***replaced***

- Dependence:
  - Height of two related individuals
  - Duration of successive eruptions of Old Faithful geyser
  - Successively drawing a king from a deck, if card from each draw is **not *replaced***

Image: https://en.wikipedia.org/wiki/Old_Faithful

# Example of independence vs. dependence

- Independence: All manufacturers have identical product mix. $p(X = x \mid Y = y) = p(X = x)$.
- Dependence: American manufacturers love SUVs, Europeans manufacturers don't.

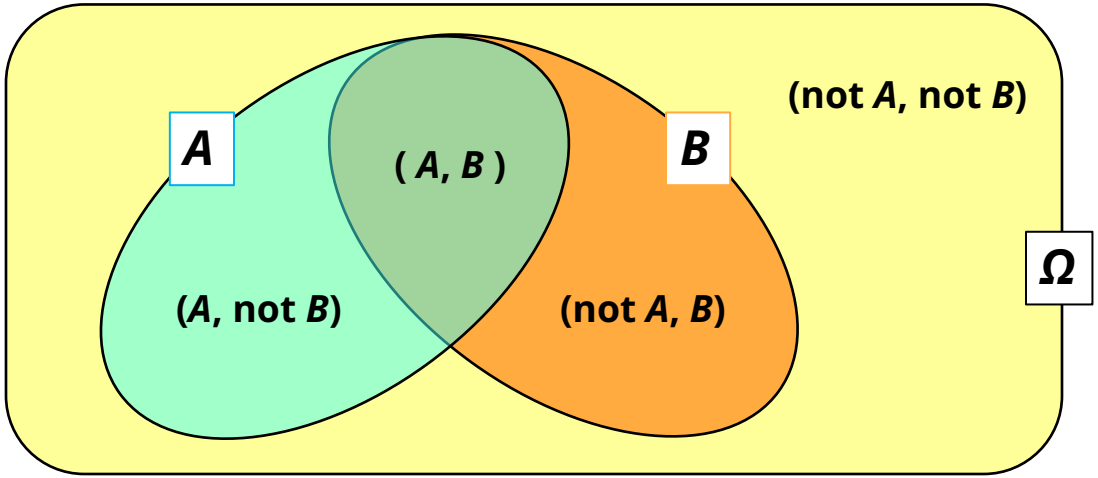


# Bayes rule

A way to find conditional probabilities for one variable when conditional probabilities for another variable are known.

$$p(B \mid A) = p(A \mid B) \cdot p(B) / p(A)$$

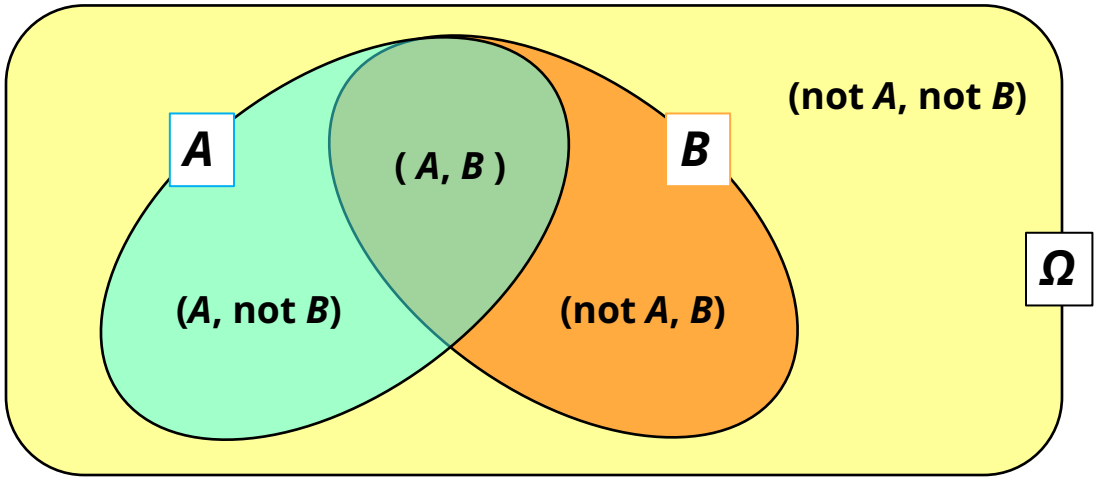
where $p(A) = p(A, B) + p(A, \text{not } B)$



# Bayes rule

posterior probability ∝ likelihood × prior probability

$$p(B \mid A) = p(A \mid B) \cdot p(B) \,/\, p(A)$$



## Example of Bayes rule

- Marie is getting married tomorrow at an outdoor ceremony in the desert. In recent years, it has rained only 5 days each year. Unfortunately, the weatherman is forecasting rain for tomorrow. When it actually rains, the weatherman has forecast rain 90% of the time. When it doesn't rain, he has forecast rain 10% of the time. What is the probability it will rain on the day of Marie's wedding?

- Event *A*: The weatherman has forecast rain.

- Event *B*: It rains.

- We know:
  - $p(B) = 5 / 365 = 0.0137$ [ It rains 5 days out of the year. ]
  - $p(\text{not } B) = 360 / 365 = 0.9863$
  - $p(A \mid B) = 0.9$ [ When it rains, the weatherman has forecast rain 90% of the time. ]
  - $p(A \mid \text{not } B) = 0.1$ [When it does not rain, the weatherman has forecast rain 10% of the time.]

# Example of Bayes rule, cont'd.

- We want to know $p(B \mid A)$, the probability it will rain on the day of Marie's wedding, given a forecast for rain by the weatherman. The answer can be determined from Bayes rule:

1. $p(B \mid A) = p(A \mid B) \cdot p(B) / p(A)$
2. $p(A) = p(A \mid B) \cdot p(B) + p(A \mid \text{not } B) \cdot p(\text{not } B) =$
   $(0.9) \cdot (0.014) + (0.1) \cdot (0.986) = 0.111$
3. $p(B \mid A) = (0.9) \cdot (0.0137) / 0.111 = 0.111$

- The result seems unintuitive but is correct. Even when the weatherman predicts rain, it only rains only about 11% of the time. Despite the weatherman's gloomy prediction, it is unlikely Marie will get rained on at her wedding.

# Probabilities: when to add, when to multiply

- **ADD**: When you want to allow for occurrence of any of several possible outcomes of a *single* process. Comparable to logical OR.

- **MULTIPLY**: When you want to allow for simultaneous occurrence of *particular* outcomes from *more than one* process. Comparable to logical AND.
  - But only if the processes are *independent.*

# Linear algebra applications

1) Operations on or between vectors and matrices
2) Coordinate transformations
3) Dimensionality reduction
4) Linear regression
5) Solution of linear systems of equations
6) Many others

Applications 1) – 4) are directly relevant to this course.

# Why vectors and matrices?

- Most common form of data organization for machine learning is a 2D array, where
  - ***rows*** represent samples (records, items, datapoints)
  - ***columns*** represent attributes (features, variables)
- Natural to think of each sample as a *vector* of attributes, and whole array as a *matrix*

vector

| Refund | Marital Status | Taxable Income | Cheat |
|---|---|---|---|
| Yes | Single | 125K | No |
| No | Married | 100K | No |
| No | Single | 70K | No |
| Yes | Married | 120K | No |
| No | Divorced | 95K | Yes |
| No | Married | 60K | No |
| Yes | Divorced | 220K | No |
| No | Single | 85K | Yes |
| No | Married | 75K | No |
| No | Single | 90K | Yes |

matrix

# Vectors

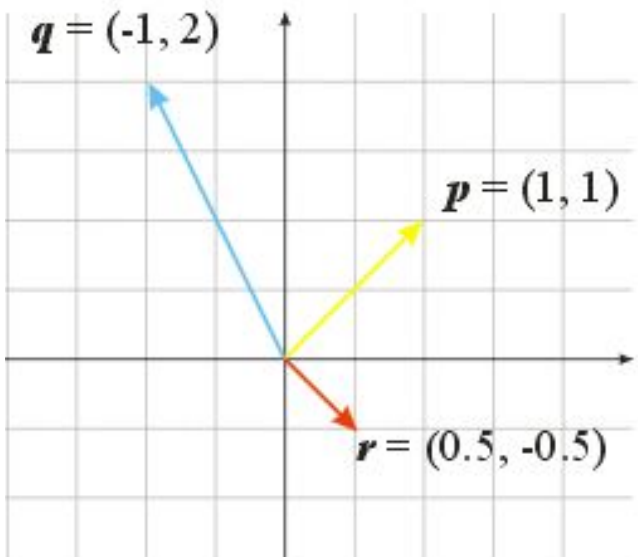
- Definition: an ***n*-tuple of values** (usually real numbers).
  - *n* referred to as the **dimension** of the vector
  - *n* can be any positive integer, from 1 to infinity
- Can be written in column form or row form
  - Column form is conventional
  - Vector elements referenced by subscript

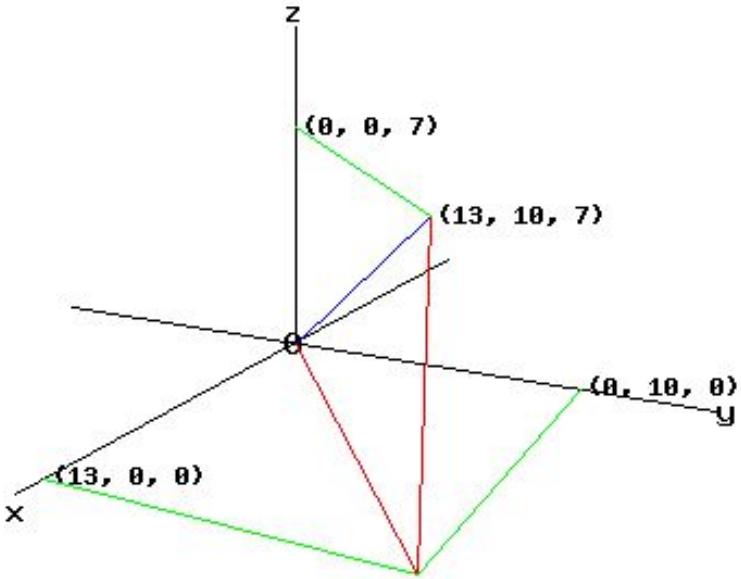
$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \qquad \mathbf{x}^{\mathrm{T}} = \begin{pmatrix} x_1 & \cdots & x_n \end{pmatrix}$$

$^{\mathrm{T}}$ means "transpose"

# Vectors

- Can think of a vector as:
  - a point in space, *or*
  - a line segment with a magnitude and direction



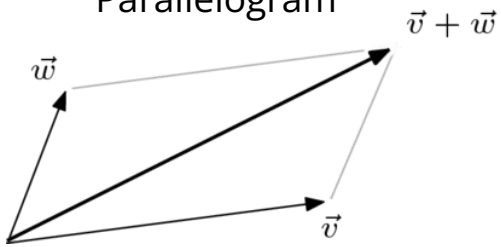
# Vector arithmetic

- Addition of two vectors
  - add corresponding elements

$$\mathbf{z} = \mathbf{x} + \mathbf{y} = \left(x_1 + y_1 \quad \cdots \quad x_n + y_n\right)^{\mathrm{T}}$$

  - result is a vector,

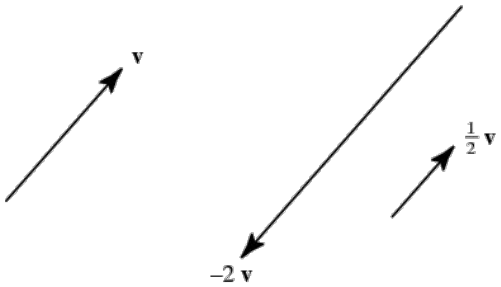
Parallelogram




- Scalar multiplication of a vector
  - multiply each element by scalar

$$\mathbf{y} = a\mathbf{x} = \left(a\,x_1 \quad \cdots \quad ax_n\right)^{\mathrm{T}}$$

  - result is a vector



# Vector arithmetic

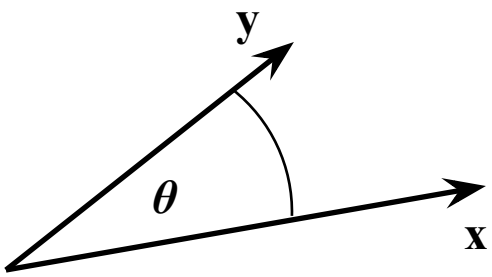
- Dot product of two vectors
  - multiply corresponding elements, then add products

$$a = \mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^{n} x_i y_i$$

  - result is a <u>scalar</u>

- Dot product alternative form

$$a = \mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos(\theta)$$



# Matrices

- Definition: an *m* x *n* 2-dimensional array of values (usually real numbers).
  - *m* rows
  - *n* columns
- Matrix referenced by 2-element subscript
  - first element in subscript is row
  - second element in subscript is column
  - example: $\mathbf{A}_{24}$ or $a_{24}$ is element in second row, fourth column of **A**

$$\mathbf{A} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$$

# Matrices

- A vector can be regarded as special case of a matrix, where one of matrix dimensions = 1.
- Matrix **transpose** (denoted $^{T}$)
  - swap columns and rows (flip about the diagonal)
    - row 1 becomes column 1, etc.
  - *m* x *n* matrix becomes *n* x *m* matrix
  - Ex:

$$\mathbf{A} = \begin{pmatrix} 2 & 7 & -1 & 0 & 3 \\ 4 & 6 & -3 & 1 & 8 \end{pmatrix} \qquad \mathbf{A}^{\mathrm{T}} = \begin{pmatrix} 2 & 4 \\ 7 & 6 \\ -1 & -3 \\ 0 & 1 \\ 3 & 8 \end{pmatrix}$$

# Matrix arithmetic

- Addition of two matrices
  - matrices must be same size
  - add corresponding elements:

    $$c_{ij} = a_{ij} + b_{ij}$$

  - result is a matrix of same size

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \begin{pmatrix} a_{11} + b_{11} & \boxtimes & a_{1n} + b_{1n} \\ \boxtimes & \boxtimes & \boxtimes \\ a_{m1} + b_{m1} & \boxtimes & a_{mn} + b_{mn} \end{pmatrix}$$

- Scalar multiplication of a matrix
  - multiply each element by scalar:

    $$b_{ij} = d \cdot a_{ij}$$

  - result is a matrix of same size

$$\mathbf{B} = d \cdot \mathbf{A} = \begin{pmatrix} d \cdot a_{11} & \boxtimes & d \cdot a_{1n} \\ \boxtimes & \boxtimes & \boxtimes \\ d \cdot a_{m1} & \boxtimes & d \cdot a_{mn} \end{pmatrix}$$

# Matrix arithmetic

- Matrix-matrix multiplication
  - vector-matrix multiplication just a special case

***TO THE BOARD!!***

- Multiplication is associative

$$\mathbf{A} \cdot (\mathbf{B} \cdot \mathbf{C}) = (\mathbf{A} \cdot \mathbf{B}) \cdot \mathbf{C}$$

- Multiplication is ***not*** commutative

$$\mathbf{A} \cdot \mathbf{B} \neq \mathbf{B} \cdot \mathbf{A} \quad \text{(generally)}$$

- Transposition rule:

$$(\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T$$

# Matrix arithmetic

- *RULE*: In any chain of matrix multiplications, the *column* dimension of one matrix in the chain must match the *row* dimension of the *following* matrix in the chain.
- Examples

$\mathbf{A}$ 3 x 5 $\mathbf{B}$ 5 x 5 $\mathbf{C}$ 3 x 1

Right:

$\mathbf{A}_{3x5} \cdot \mathbf{B}_{5x5} \cdot (\mathbf{A}^T)_{5x3}$

$\mathbf{C}^T \cdot \mathbf{A} \cdot \mathbf{B}$ $\mathbf{A}^T \cdot \mathbf{A} \cdot \mathbf{B}$ $\mathbf{C} \cdot \mathbf{C}^T \cdot \mathbf{A}$

Wrong:

$\mathbf{A} \cdot \mathbf{B} \cdot \mathbf{A}$ $\mathbf{C} \cdot \mathbf{A} \cdot \mathbf{B}$ $\mathbf{A} \cdot \mathbf{A}^T \cdot \mathbf{B}$ $\mathbf{C}^T \cdot \mathbf{C} \cdot \mathbf{A}$

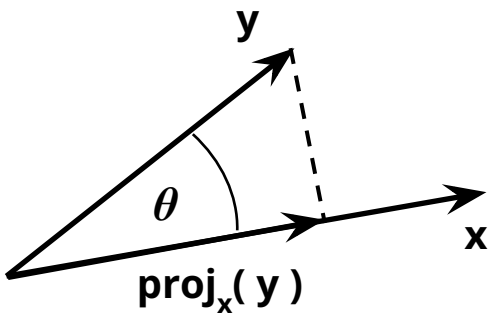
# Vector projection

- Orthogonal projection of **y** onto **x**
  - Can take place in any space of dimensionality $\geq 2$
  - Unit vector in direction of **x** is

    $$\mathbf{x} / ||\mathbf{x}||$$

  - Length of projection of **y** in direction of **x** is

    $$||\mathbf{y}|| \cdot \cos(\theta)$$




  - Orthogonal projection of **y** onto **x** is the vector

    $$\mathbf{proj}_{\mathbf{x}}(\mathbf{y}) = \mathbf{x} \cdot ||\mathbf{y}|| \cdot \cos(\theta) / ||\mathbf{x}|| =$$

    $$[(\mathbf{x} \cdot \mathbf{y}) / ||\mathbf{x}||^2]\,\mathbf{x}$$ (using dot product alternate form)

## Optimization theory topics

- Maximum likelihood
- Expectation maximization
- Gradient descent